MEDIUM

GEOMETRY AND TRIGONOMETRY

Circles

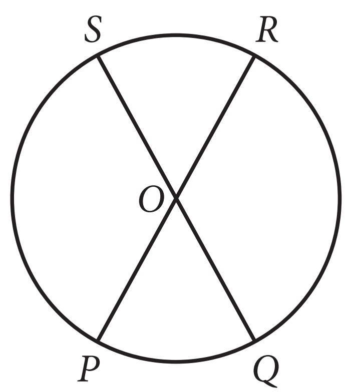
01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- The center of the circle is point upper O.
- Points upper S, upper R, upper Q, and upper P are on the circle.
- Line segment upper P upper R is a diameter of the circle.
- Line segment upper Q upper S is a diameter of the circle.
- Diameters upper P upper R and upper Q upper S intersect at point upper O.
- A note indicates the figure is not drawn to scale.

The circle shown has center O, circumference 144π, and diameters $\overline{PR}$ and $\overline{QS}$. The length of arc PS is twice the length of arc PQ. What is the length of arc QR?

A. 24π

B. 48π

C. 72π

D. 96π

Q2 GEOMETRY AND TRIGONOMETRY

The measure of angle Z is $60°$. What is the measure, in radians, of angle Z?

A. $\frac{1}{6}\pi$

B. $\frac{1}{3}\pi$

C. $\frac{2}{3}\pi$

D. 1π

Q3 GRID-IN GEOMETRY AND TRIGONOMETRY

What is the radius of the circle in the *xy*-plane defined by $x + 2^2 + y + 5^2 = 169$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4

GRID-IN GEOMETRY AND TRIGONOMETRY

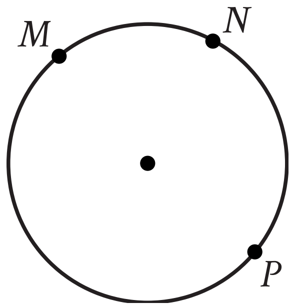
An angle has a measure of $\frac{9\pi}{20}$ radians. What is the measure of the angle in degrees?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q5 GEOMETRY AND TRIGONOMETRY



- Clockwise from top left, the following points are on the circle:
  - upper M
  - upper N
  - upper P

Points M, N, and P lie on the circle shown. On this circle, minor arc MN has a length of 39 centimeters and major arc MPN has a length of 195 centimeters. What is the circumference, in centimeters, of the circle shown?

A. 39

B. 156

C. 195

D. 234

Q6 GEOMETRY AND TRIGONOMETRY

A circle in the xy-plane has its center at -4, 5 and the point -8, 8 lies on the circle. Which equation represents this circle?

A. $x - 4^2 + y + 5^2 = 5$

B. $x + 4^2 + y - 5^2 = 5$

C. $x - 4^2 + y + 5^2 = 25$

D. $x + 4^2 + y - 5^2 = 25$

Q7 GEOMETRY AND TRIGONOMETRY

In the xy-plane, a circle with radius 5 has center $(-8,6)$. Which of the following is an equation of the circle?

A. $(x-8)^2+(y+6)^2=25$

B. $(x+8)^2+(y-6)^2=25$

C. $(x-8)^2+(y+6)^2=5$

D. $(x+8)^2+(y-6)^2=5$

Q8

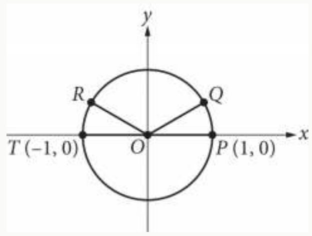
GRID-IN GEOMETRY AND TRIGONOMETRY

The number of radians in a 720-degree angle can be written as $a\pi$, where a is a constant. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.



In the xy-plane above, points P, Q, R, and T lie on the circle with center O. The degree measures of angles POQ and ROT are each 30°. What is the radian measure of angle QOR ?

A. $\frac{5}{6}\pi$

B. $\frac{3}{4}\pi$

C. $\frac{2}{3}\pi$

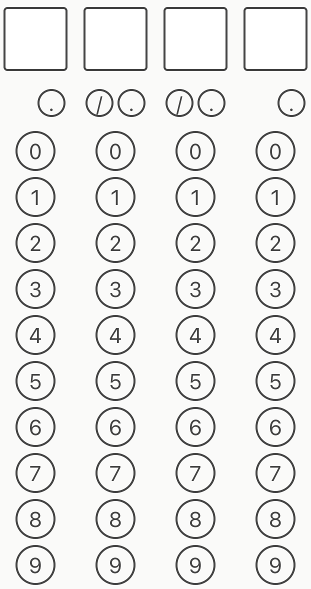
D. $\frac{1}{3}\pi$

Q10

GRID-IN GEOMETRY AND TRIGONOMETRY

An angle has a measure of $\frac{16\pi}{15}$ radians. What is the measure of the angle, in <u>degrees</u>?

STUDENT-PRODUCED RESPONSE



Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.